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The effects of gamification on students' emotions

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ABSTRACT

Gamification is widely used in educational systems to enhance engagement and learning experiences. However, its effects on learners' emotions remain underexplored. This study investigates the impact of gamification on students' basic emotions and emotional transitions during interaction with a learning management system (LMS). A controlled between-subjects experiment was conducted with 62 university students using either a gamified or non-gamified LMS. Emotional responses were captured in real time through automated facial-expression recognition and analyzed by duration and frequency. Anger was the most frequent emotion and fear the least frequent in both conditions. Statistical analyses found no significant differences between groups in emotion duration or frequency. However, emotional transition analyses revealed distinct patterns: anger-to-happiness was the most common transition in the gamified condition, whereas disgust-to-anger predominated in the non-gamified condition. These findings suggest that the specific gamification design used in this study did not affect basic emotional expressions but may have influenced emotional dynamics during learning. The study contributes objective evidence on emotions in gamified learning environments and highlights the importance of examining emotional transitions beyond static emotional states.

Introduction

Gamification, i.e., "the design of systems, services, and activities to provide motivational benefits similar to those games usually create"[1,2], has garnered significant interest from researchers and practitioners as a strategy for promoting positive behavioral change[3,4]. In learning systems, gamification has been widely adopted to enhance user engagement, motivation, and learning outcomes[5,6]. By incorporating gameful dynamics such as acknowledgment, storytelling, and narratives, designers and educators aim to create more interactive and enjoyable learning experiences, potentially fostering positive emotions and improving learning outcomes[7,8].

Despite the widespread adoption of gamification in learning systems and the promising results reported in various studies[9], the link between engaging in learning with a gamified system and the learners' emotional states has seen limited attention in academia[10]. Past research appears to have focused more on learners' cognitive and behavioral outcomes, and less on emotional outcomes[11,12], which limits the understanding of how gamified environments affect students' emotions[13]. Understanding how gamification influences students' basic emotions, *e.g.*, happiness, sadness, anger, surprise, fear, and disgust[14], is crucial[15,16], as emotions play a significant role in learning and motivation to learn[17,18].

Addressing this gap, we investigated the effects of gamification on students' basic emotions (i.e., happiness, sadness, anger, surprise, fear, and disgust) and emotional transitions, thus answering the following research questions: **RQ1**: Does gamification affect the total duration of students' emotions during LMS interaction? **RQ2**: Does gamification affect the frequency of students' emotions during LMS interaction? **RQ3**: What are the sequential relationships between students' emotions during LMS interaction? These RQs compare emotional duration and frequency across gamified and non-gamified LMS versions and identify emotional sequences by examining which emotions precede others and how these patterns differ between conditions. These insights can support the design of emotionally supportive gamified learning environments.

To answer these research questions, this study employed a controlled between-subjects experimental design involving 62 university students who interacted with either a gamified or a non-gamified version of the same LMS. Participants completed identical learning tasks while their emotional responses were continuously monitored using automated facial-expression recognition software. Six basic emotions (happiness, sadness, anger, surprise, fear, and disgust) were analyzed in terms of duration, frequency, and transition patterns. Statistical analyses were then conducted to compare emotional experiences across conditions and to identify the most frequent emotional sequences during the learning process.

Our findings show that anger was the most commonly observed emotion across both system versions, while fear was the least expressed in both. However, the analysis revealed no significant differences in emotional expressions between the

gamified and non-gamified groups. The results also indicate that in the experimental group, the most frequent transition occurred from angry to happy. In the control group, the most frequent transition occurred from disgusted to angry. These results contribute to learning technology and gamification research by extending existing emotion-focused studies with objective, automated measures of facial expressions collected during interaction with a learning management system. Furthermore, by examining emotional transition patterns alongside emotional frequency and duration, the study provides additional insights into the temporal dynamics of students' emotional experiences that are often overlooked in traditional emotion assessments.

Results

Descriptive analysis

To initially characterize our sample and facilitate subsequent group comparisons, we calculated descriptive statistics, including measures of central tendency (i.e., mean) and variability/variance (VAR) and standard deviation (SD) for the students' emotions in both experimental (gamified) and control (non-gamified) groups. Figure 1 presents the average total time for each emotion besides neutral, removed to provide better visibility to the other values.

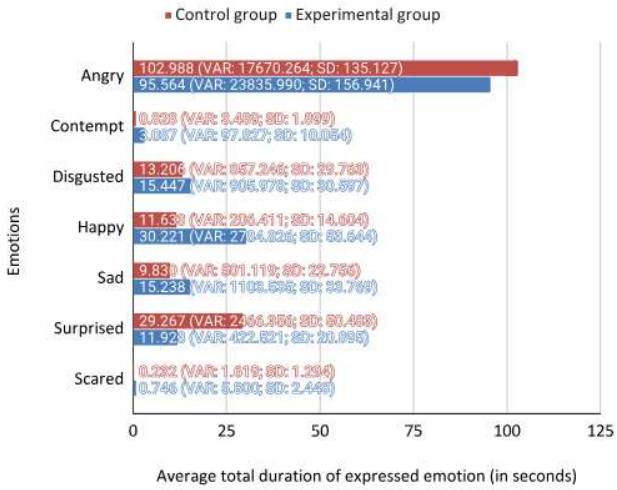


Figure 1. Average total duration of expressed emotion (in seconds) for each emotion in both groups.

The results indicate that, most of the time, participants displayed neutral facial expressions across both versions of the system. Moving forward, anger was the most frequently observed emotion in both system versions, whereas fear was the least expressed emotion in both versions. We observed that the control group expressed more angry and surprised emotions than the experimental group, which expressed more contempt, disgust, happy, sad, and scared emotions. Furthermore, anger was the primary emotion, followed by happiness in the experimental group and surprise in the control group.

RQ1: Does gamification affect the total duration of students' emotions during LMS interaction?

The results of the inferential analysis demonstrated that there was no significant difference in the participants' emotions between the groups: Neutral ($t = -0.511$, $p = .611$); Angry ($U = 488.5$, $Z = 0.25$, $p = .91$); Contemp ($U = 457.0$, $Z = -0.75$, $p = .68$); Disgusted ($U = 433.0$, $Z = -1.52$, $p = .47$); Happy ($U = 419.5$, $Z = -1.95$, $p = .39$); Sad ($U = 417.5$, $Z = -2.01$, $p = .36$); Surprised ($U = 531.0$, $Z = 1.61$, $p = .47$); and Scared ($U = 449.0$, $Z = -1.00$, $p = .30$). The associated effect size was small for both emotions: Neutral ($d = .13$); Angry ($r = .03$); Contemp ($r = .10$); Disgusted ($r = .19$); Happy ($r = .25$); Sad ($r = .26$); Surprised ($r = .20$); and Scared ($r = .13$). The result suggests a limited practical impact of gamification on emotions in the context of this study[19]. Table 1 summarizes the results.

In summary, our findings indicate no statistically significant differences between groups for any measured emotions. Associated effect sizes were also small across all emotions, suggesting a limited practical impact of the gamification intervention on students' emotional experiences in this specific context. These findings contribute to the existing literature by indicating that the gamification design employed in this study did not produce detectable changes in basic emotions during the experimental session. However, the absence of statistically significant differences should not be interpreted as evidence that gamification has no emotional effects more generally, as other implementations, contexts, levels of exposure, or learner populations may yield different outcomes.

Table 1. Inferential analysis (average time).

Group	N	M	S	t or U	Z	p	*d* or *r*
Neutral C	31	1151.711	319.486	-0.511	-	.611	.13
Neutral E	31	1188.844	248.105				
Angry C	31	31.758	984.500	488.500	0.256	.915	.03
Angry E	31	31.241	968.500				
Contemp C	31	30.741	953.000	457.000	-0.752	.680	.10
Contemp E	31	32.258	1000.000				
Disgusted C	31	29.96	929.000	433.000	-1.520	.479	.19
Disgusted E	31	33.032	1024.000				
Happy C	31	29.532	915.500	419.500	-1.952	.391	.25
Happy E	31	33.467	1037.500				
Sad C	31	29.467	913.500	417.500	-2.016	.366	.26
Sad E	31	33.532	1039.500				
Surprised C	31	33.129	1027.000	531.000	1.616	.478	.21
Surprised E	31	29.870	926.000				
Scared C	31	30.483	945.000	449.000	-1.008	.305	.13
Scared E	31	32.516	1008.000				

Key: The line highlighted in light gray shows the t-test result (the remaining lines show the results of the Mann-Whitney U test). N: Number of participants; M: Mean (for t-test) or Mean rank (for Mann-Whitney U test); S: Std. Dev. (for t-test) or Sum of ranks (for Mann-Whitney U test); *t*: t-test t; U: Mann-Whitney U; Z: Z score; p: p-value; *d*: Cohen's *d* (for t-test effects size); *r*: r-value (for Mann-Whitney effect size); C: Control group; E Experimental group.

RQ2: Does gamification affect the frequency of students' emotions during LMS interaction?

The results of the inferential analysis demonstrated that there was no significant difference in the frequency of participants' emotions between the groups: Neutral ($U = 554.500$, $Z = 2.368$, $p = .301$); Angry ($U = 490.000$, $Z = 0.304$, $p = .899$); Contemp ($U = 452.500$, $Z = -0.896$, $p = .622$); Disgusted ($U = 425.500$, $Z = -1.760$, $p = .412$); Happy ($U = 413.500$, $Z = -2.144$, $p = .345$); Sad ($U = 413.500$, $Z = -2.144$, $p = .335$); Surprised ($U = 516.000$, $Z = 1.136$, $p = .618$); Scared ($U = 449.500$, $Z = -0.992$, $p = .313$). The associated effect size was small for both emotions: Neutral ($r = .30$); Angry ($r = .04$); Contemp ($r = .11$); Disgusted ($r = .22$); Happy ($r = .27$); Sad ($r = .27$); Surprised ($r = .14$); Scared ($r = .13$) The result suggests a limited practical impact of gamification on emotions in the context of this study[19]. Table 2 summarizes the results.

In summary, in the same direction as our first research question, our findings indicate no statistically significant differences in the frequency of the measured emotions between the two groups. Associated effect sizes were also small across all emotions, suggesting a limited practical impact of the gamification intervention on students' emotional experiences in this specific context. These findings suggest that the gamification design adopted in this experiment did not produce detectable differences in the frequency of students' emotional expressions. Therefore, our results should be interpreted as evidence regarding this particular implementation rather than as evidence against the emotional effectiveness of gamification as a broader design approach.

RQ3: What are the sequential relationships between students' emotions during LMS interaction?

Given the exploratory nature of the emotional transition analysis and the relatively limited sample size available for sequential analyses, transitions were examined descriptively rather than through inferential statistical modeling. Consequently, the reported transition frequencies should be interpreted as preliminary indications of potential emotional dynamics rather than statistically validated differences between conditions. Future studies with larger samples should apply transition probability models, sequential analyses, or Markov-based approaches to determine whether the observed emotional pathways differ significantly across learning environments.

An initial analysis of the transitions between emotions experienced by participants in the experimental group reveals distinct, repeating transition patterns. The most frequent transition occurred from angry to happy (19 times), indicating that episodes classified as anger were frequently followed by expressions classified as happiness during system interaction. Although this pattern may reflect emotional shifts occurring while participants interacted with the system, the present data do not allow conclusions regarding the causes underlying these transitions.

Table 2. Inferential analysis (frequency).

Group	N	M	S	t or U	Z	p	*d* or *r*
Neutral C	31	75.935	37.738	554.500	2.368	.301	.30
Neutral E	31	64.221	27.402				
Angry C	31	28.065	30.721	490.000	0.304	.899	.04
Angry E	31	21.506	18.720				
Contemp C	31	0.548	1.150	452.500	-0.896	.622	.11
Contemp E	31	0.942	1.949				
Disgusted C	31	3.944	7.254	425.500	-1.760	.412	.22
Disgusted E	31	5.149	7.739				
Happy C	31	5.355	6.902	413.500	-2.144	.345	.27
Happy E	31	7.025	8.350				
Sad C	31	3.343	6.173	413.500	-2144	.335	.27
Sad E	31	5.375	8.171				
Surprised C	31	9.663	14.210	516.000	1.136	.618	.14
Surprised E	31	7.129	10.187				
Scared C	31	0.011	0.061	449.500	-0.992	.313	.13
Scared E	31	0.033	0.102				

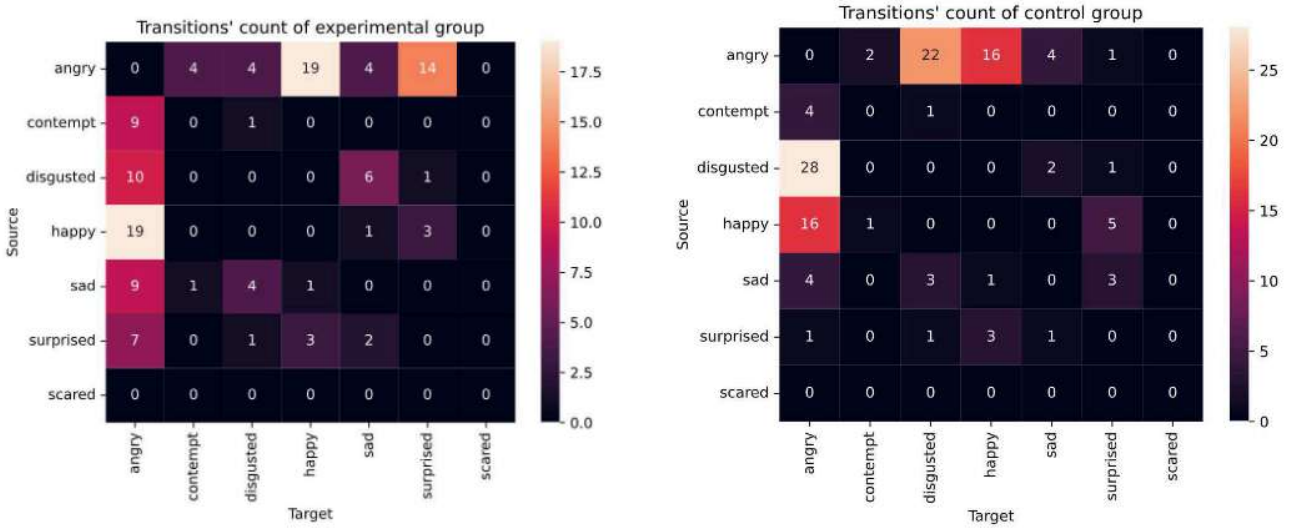
Key: N: Number of participants; M: Mean (for t-test) or Mean rank (for Mann-Whitney U test); S: Std. Dev. (for t-test) or Sum of ranks (for Mann-Whitney U test); *t*: t-test t; U: Mann-Whitney U; Z: Z score; p: p-value; *d*: Cohen's *d* (for t-test effects size); *r*: r-value (for Mann-Whitney effect size); C: Control group; E Experimental group.

Similarly, there were many transitions from angry to surprised (14 times). Since anger was the most frequently appearing emotion, it is unsurprising that we found multiple instances where participants switched emotional states from anger. Among bidirectional shifts between emotions, the most often appearing one was from happy to angry, which occurred 19 times. The transition from disgusted to happy appeared six times, reflecting a moderate shift from negative to positive emotion. Contemp and disgust were relatively stable emotions, with Contemp most frequently transitioning to anger (nine times) and disgust transitioning to anger 10 times. Notably, transitions involving scared were absent, indicating that this emotion was neither significantly experienced nor transitioned to/from other states. The surprised state was reached from angry (four times), disgusted (once), happy (three times), and sad (twice). However, it did not frequently transition further. Figure 2a presents the results.

On the other hand, analyzing the control group, the results indicate the most frequent transition occurs from "disgusted" to "angry" with a count of 28, suggesting a strong shift between these emotions. "Angry" to "disgusted" (22) and "angry" to "happy" (16) are also prominent transitions, indicating variability in emotional shifts from anger and may suggest that negative-to-negative emotional transitions appear more frequently in the non-gamified condition, possibly due to them alleviating frustration. "Happy" to "angry" (16) is another notable transition, highlighting a bidirectional shift between happiness and anger. "Sad" to "surprised" (3) and "happy" to "surprised" (5) indicate some movement toward surprise, however, with lower frequency. Minimal transitions are observed for emotions like "scared", which has no transitions in this dataset. Figure 2b presents the results.

Discussion

In an effort to fill a gap in the field of gamification in education, which is to understand how gamification affects learners' emotions, in this study, we analyzed the effects of gamification on students' emotions. Our primary finding was the absence of statistically significant differences in the measured emotional expressions between participants interacting with the gamified and non-gamified versions of the LMS. Importantly, this finding should be interpreted as the absence of detectable differences under the conditions of this study rather than definitive evidence that gamification does not affect learners' emotions. The statistical interpretation of the findings should consider the sensitivity of the study design. While the sample size was adequate for detecting large between-group differences, smaller emotional effects commonly observed in educational technology interventions may not have been identified. Accordingly, the present findings provide stronger evidence against the existence of large emotional effects than against the existence of more subtle emotional influences of gamification.



(a) Emotional transitions in the experimental group

(b) Emotional transitions in the control group

Figure 2. Emotional transitions across groups: (a) experimental; (b) control.

An important observation is that anger emerged as the most frequently detected emotional state across both experimental conditions. While this finding may indicate moments of frustration associated with task completion, alternative explanations should also be considered. Previous research has suggested that automated facial-expression recognition systems may occasionally classify facial configurations associated with concentration, cognitive effort, problem-solving, or task-focused attention as anger-like expressions[20]. Therefore, the high prevalence of anger observed in this study should not necessarily be interpreted as evidence that participants were predominantly experiencing negative affect. Instead, it may reflect a combination of genuine frustration, increased cognitive effort, and limitations inherent to facial-expression-based emotion detection. Future studies combining facial analysis with self-report measures or physiological sensors could help clarify the nature of these emotional states.

One possible explanation for the lack of significant emotional differences is the design of the gamification elements themselves. It is crucial to assess whether these features were sufficiently engaging or if participants perceived them as superficial[2]. The effectiveness of gamification largely depends on the quality and relevance of the game mechanics used[4]. If the elements did not resonate with users or align with their intrinsic motivations, they may not have elicited the intended emotional responses.

An important consideration when interpreting these findings is that the present study did not include a direct manipulation check assessing participants' perceptions of gamefulness, engagement with the gamification elements, or perceived differences between the two system versions. Consequently, although the gamified version included ten gamification elements, we cannot determine the extent to which participants actively perceived or engaged with those features. Therefore, the observed null findings may reflect either a genuinely limited emotional impact of the implemented gamification design or insufficient differentiation between the subjective experiences produced by the two versions of the system.

Another factor to consider is the duration of the experiment. The time participants spent interacting with the gamified system may not have been sufficient to produce noticeable emotional changes. Emotions are transient and influenced by various factors[21], which may include gamified elements. A longer-term study could reveal different emotional response patterns as participants become more accustomed to the gamified environment.

The nature of the educational tasks used in the study may also have played a role. If the tasks were not sufficiently challenging or engaging, they might not have elicited strong emotional reactions, regardless of the gamification[22]. Ensuring that tasks align with participants' interests and skill levels is critical in determining the emotional impact of gamification. Additionally, individual differences among participants, such as prior experiences with gamification, personal preferences, and emotional baseline, could have influenced the results[23–25]. Some participants may have been more receptive to gamification than others, potentially diluting the overall emotional impact when averaged across the entire sample.

The results reveal recurring emotional transition patterns, with certain emotional states appearing sequentially more frequently than others (see Figures 2a and 2b). A notable observation is the bidirectional transition between anger and happiness, with 19 instances of participants shifting from anger to happiness and vice versa. This finding indicates that anger and

happiness often appeared sequentially during interaction with the learning management system. While this pattern may reflect shifts between negative and positive affective states, the current study does not allow conclusions regarding the underlying psychological mechanisms responsible for these transitions.

Additionally, anger frequently transitions to surprise (14 times), indicating that unexpected events or system features may play a role in modulating emotions. One possible interpretation is that these transitions were associated with unexpected events occurring during system interaction. However, the present study was not designed to identify which specific features or events triggered individual emotional transitions. Conversely, surprise does not show a strong return transition to anger, implying that participants do not often react negatively to unexpected events. However, may process them differently.

Negative emotions such as contemp and disgust show relatively stable patterns. Contemp transitions to anger nine times, and disgust transitions to anger 10 times, highlighting a tendency for these negative states to escalate into frustration. However, disgust also transitions to happiness six times, suggesting that participants sometimes overcome their initial aversion or dissatisfaction, although the specific factors associated with these transitions cannot be determined from the current data. Interestingly, the emotion "scared" does not appear as a significant transition point, either as a source or target. This finding suggests that facial expressions classified as fear were rarely detected. This could be an important insight for refining the system, especially if inducing a sense of thrill or challenge was intended.

The relatively frequent transition from sadness to anger (nine times) also suggests that frustration may follow moments of discouragement, potentially when participants struggle with challenges. However, sadness does not transition to happiness at a high rate, indicating that overcoming sadness may require additional system mechanisms beyond those currently in place. The prominence of transitions involving anger, happiness, and surprise suggests that these emotional states frequently co-occurred within participants' emotional trajectories during system interaction. However, no conclusions can be drawn regarding engagement levels based solely on these transition frequencies.

Threats to validity and limitations

This study identified several threats to validity and limitations, each with potential implications for the interpretation of our findings. Although measures were taken to mitigate these challenges, some limitations remain inherent to the study design. Initially, while participants were randomly assigned to the experimental and control groups, the manual method of turning a cup for group assignment could introduce bias or errors. To mitigate this, randomization was conducted immediately before the experiment in the presence of two researchers, minimizing the potential for systematic bias.

We relied on Noldus Observer XT and FaceReader for detecting and analyzing emotional responses. Although these tools are widely regarded for their accuracy and reliability, any inherent inaccuracies could affect the results. To address this, all equipment was calibrated and tested according to the software manuals before use. At the same time, potential confounders, such as prior experience with gamified systems, individual motivation levels, or familiarity with the educational content, were not controlled. Standardized tasks and uniform instructions aimed to minimize these effects. However, residual influence remains a possibility.

Another limitation concerns treatment fidelity. Although participants in the experimental condition interacted with a version of the LMS containing gamification elements, the present study did not directly assess the extent to which participants noticed, used, or actively engaged with individual gamification features. Consequently, we cannot determine whether all participants similarly experienced the gamified environment. Future studies should incorporate system log data and behavioral interaction metrics to investigate participants' engagement with specific gamification elements and their relationship with emotional responses.

An additional consideration concerns the interpretation of facial-expression recognition data as indicators of emotional states. Although FaceReader is a widely used and validated tool for automated affect detection, the broader literature contains ongoing debate regarding the extent to which emotional experiences can be reliably inferred from facial expressions alone. While basic-emotion theories suggest that facial movements can provide meaningful indicators of underlying emotions, alternative perspectives argue that emotions are context-dependent constructions that may not map consistently onto specific facial configurations. Consequently, the emotional measurements reported in this study should be interpreted as estimates derived from observable facial-expression patterns rather than as direct measurements of participants' subjective emotional experiences.

Another limitation is that the present study did not directly assess learning outcomes, focusing instead on emotional experiences during system interaction. Consequently, it is not possible within this specific study to determine whether the observed emotional patterns were associated with differences in learning performance. Future research should continue to investigate the relationship between emotional dynamics and educational outcomes using integrated analytical approaches.

The sample size of 62 participants provided sufficient statistical sensitivity to detect large effects but offered more limited power for identifying small or moderate effects. Consequently, while the analyses do not support the existence of large emotional differences between conditions, smaller effects may have remained undetected. Therefore, the absence of statistically

significant differences should not be interpreted as evidence of equivalence between the gamified and non-gamified conditions. Future studies using larger samples would enable more precise estimation of potential emotional effects and increase confidence in null findings. In the same line, our sample consisted of students from a European university, representing 19 countries. While this diversity adds value, it limits the applicability of findings to other educational contexts or populations. Future research should replicate the study with more varied demographics and settings to enhance external validity.

Conducting the study in a controlled laboratory environment allowed precise control over experimental conditions. However, it may not reflect real-world educational settings. Field studies in authentic environments are recommended for future research to validate these findings. Moreover, participants interacted with the system for up to 30 minutes, which may not capture the long-term effects of gamification on emotions. Future research should investigate extended interaction periods to better understand the sustained impact of gamified systems.

The use of standardized tasks ensured control over task-related variables. However, it reduced the generalizability of diverse educational activities. Including a broader range of tasks in future studies would improve the applicability of findings. Similarly, the use of observational tools to measure emotional responses might introduce bias if participants were aware of being observed. To mitigate this, participants were not informed about the focus on their emotional responses, minimizing behavioral alterations due to observation.

Despite the Eagle-edu system's prior use in various studies, participants' familiarity or comfort with the platform could introduce biases. To address this, a standardized introduction to the system was provided to all participants, reducing unfamiliarity. Including participants from 19 countries introduces variability in emotional responses due to cultural differences in emotional expression. While this diversity enriches the study, it complicates result interpretation. Future studies should explicitly consider cultural factors and potentially analyze data for separate cultural groups.

Finally, regarding the analysis of emotional transitions, due to the available sample size and the comparatively low occurrence of several emotional states, transition analyses were conducted descriptively. Therefore, although the observed patterns provide useful insights into potential emotional dynamics during system interaction, they should be interpreted cautiously and should not be viewed as evidence of statistically significant differences in emotional trajectories between the experimental conditions.

Theoretical implications

This study advances our understanding of the outcomes of engaging with gamified learning tools and supports the existing research that has focused on cognitive and behavioral outcomes[11,12,26] by including the perspective of emotional outcomes. We provided empirical evidence on the impacts of one gamified learning tool on the basic emotions[14] of happiness, sadness, anger, surprise, fear, and disgust in a controlled laboratory setting, which contributes to the growing body of research investigating emotions in gamified environments through objective measurement approaches. In our empirical study, there were no statistically significant differences in experienced emotions between a group using a standard version of the tool and a group using a gamified version of the tool. Despite this, we saw multiple different emotions manifest rather consistently across the participant pool. This finding indicates that the gamification design implemented in the present study may not play as significant a role as could be assumed based on previous research (e.g.,[27–29]). However, it remains unclear whether we would see larger differences with heavier gamification approaches or games that build on elements such as fictional narratives, world building, social challenges, and immersive gameplay[3,22,30].

The present study deliberately focused on emotional processes rather than learning outcomes. This decision was motivated by the specific objective of examining how students' emotional states evolve during interaction with a gamified learning environment and by the existing emphasis of the gamification literature on cognitive and performance-related outcomes[31]. Importantly, learning outcomes were investigated separately within the broader research project from which this study emerged. Previous work using the same experimental context identified positive effects of gamification on students' academic performance, suggesting that improvements in learning outcomes may occur even in the absence of substantial differences in basic emotional expressions. Taken together, these findings highlight the importance of examining emotional and learning-related outcomes as complementary, but not necessarily equivalent, dimensions of the gamified learning experience.

The use of automated emotion detection in this study aligns with the broader goals of affective computing within smart educational contexts. By leveraging objective, real-time data from FaceReader and eye-tracking, this research moves beyond the limitations of subjective self-reports to provide a high-resolution map of the learner's emotional state. These findings contribute to the theoretical understanding of how 'smart' systems can perceive human affect, suggesting that while gamification elements are identifiable by the system, they do not fundamentally alter the primary emotional intensity elicited by the instructional content itself. This underscores a need for theoretical models in smart education that prioritize the interplay between automated physiological sensing and core task design.

According to the cognitive-emotional view on gamification by Mullins and Sabherwal[27], the outcomes of a gameful system are always the result of an interplay between cognitions and emotions. Thus, by shedding light on the emotional side of gameful

learning tools, we provide new empirical evidence to support existing theoretical models of gamification outcomes, such as the cognitive-emotional view on gamification[27]. More precisely, our findings suggest that the core learning content may be a key driver of student emotions, and while gamification has some impact, this impact is small in comparison. We also contribute to this body of literature through empirical evidence on transitions between emotions while engaged with a gamified learning system. We show that there are both bidirectional and unidirectional relationships between basic emotions, and this opens the need for more research into the shifting emotional states and their relationship with learning. These findings relate to work on the importance of linking learning attributes to game mechanics, a crucial link that has been addressed in depth in previous research[32–34]. In light of this body of literature, it is also important to examine the quality of the gamification implementation concerning the observed student emotions.

Another possible explanation for the lack of significant emotional differences between gamified and non-gamified conditions is through the Self-Determination Theory, where motivation and engagement are driven by the core psychological needs of autonomy, competence, and relatedness[35]. The gamification elements may have been perceived as rigid or externally imposed, which would make students feel the game is controlling instead of motivating, reducing internal motivation, and increasing neutral responses. This may be mitigated by adaptive gamification elements that allow students to select rewards and set personal goals. They may have also felt little sense of progress due to the lack of more immediate and clear feedback (high levels of anger to surprised and angry to happy in the experimental group, in Figure 2a), and a possible lack of achievement (frequently transitioning from happy to angry in Figure 2a). Adding more challenge-based gamification elements, scaffolded difficulty in our questionnaires, and more meaningful feedback may help increase competence-driven engagement. Finally, happiness and surprise may increase in the experimental group with an increase in relatedness by adding more socially related gamification elements like team-based challenges and cooperative quests[36].

The lack of significant emotional differentiation between the gamified and non-gamified groups suggests that 'meta-game' interactions, typical of many current smart platforms, may lack the depth required to influence basic achievement emotions. Theoretically, this indicates that for gamification to be effective in an interactive smart environment, it must be 'intrinsic', embedded directly into the learning mechanics rather than added as a reward layer. This finding encourages a shift in the design of interactive technology away from extrinsic motivators, like leaderboards and points, toward smart instructional designs that gamify the cognitive challenge itself, thereby fostering a more authentic and emotionally resonant interactive experience.

Practical implications

Our work provides two practical implications for gamified learning system design. First, since in our empirical context gamification appeared to have a non-significant impact on basic emotions, designers could focus more on other aspects, such as cognitive load management and meaningfulness of the learning tasks, to foster positive emotions in participants. Of course, gamification is a broad umbrella term for various solutions, and our empirical findings offer only perspectives on one context, so we encourage practitioners to apply these findings through trial, experimentation, and testing. Second, the discovered transitions between emotional states could be more knowingly manipulated, and designers could even look forward to mapping emotional state pathways for learners where key learning content is tied to moments of positive emotions. Designers could also examine system interaction logs alongside emotional data to identify which gameful features are associated with specific emotional trajectories. Such information may help designers refine gamification implementations and better understand how learners engage with individual elements throughout the learning process.

A key practical contribution for designers of smart learning environments is the identification of specific emotional transition patterns, such as the frequent shift from anger to happiness observed in this study. In a truly 'smart' system, these detected transitions can serve as algorithmic triggers for adaptive interventions. For instance, if an automated system detects that a student is lingering in a state of frustration (anger) without transitioning to a positive state (happiness), it could proactively adjust the difficulty level or provide scaffolded feedback. Utilizing these objective transition matrices allows developers to move toward 'emotionally-aware' interfaces that respond dynamically to the user's real-time psychological needs rather than relying on static gamification wrappers.

Our work also provides some insights into UX research practices at companies beyond the realm of gamification. Today, multiple companies engage in UX research efforts to better understand how various product design elements are likely to be received, or are actively received, by the target audience[37–39]. We demonstrate that we can observe differences in emotions between participants using a gameful system and a non-gamified version of it. Such data could be useful for companies carrying out A/B testing to identify which version is better received by target audiences[40], or help UX researchers identify user pain points[41]. While our study was conducted in a controlled laboratory environment, similar studies could potentially be done remotely as well, with specific instructions provided to participants regarding how they should set up a camera to film themselves. Such studies can offer game and learning environment UX researchers new tools for understanding their participants' experiences.

Research agenda

For future research, we encourage replicating this work with other learning systems to better understand what context-dependent phenomena may have appeared in our data. For example, it could be that the learning system we used provokes an unusually high amount of anger in participants, and that the basic emotions experienced with a competing system look different. Such studies would add more robustness to our findings and contribute important additional empirical evidence for building more generalizable theoretical frameworks for understanding the impacts of gamification on learners' emotions in an educational setting.

We also encourage future studies to map specific emotions to specific instances in learning scenarios and to examine participants' interactions with individual gamification elements through system logs and behavioral analytics. Such analyses would enable researchers to investigate treatment fidelity and identify which gamification features are actually perceived and engaged with by learners, thereby improving our understanding of how specific gameful affordances relate to emotional experiences.

Another avenue for future research relates to the way emotions are detected. According to[42], the affective neuroscience approach, emotions are fundamental and shared across mammalian species. There are circuits in our brain specifically responsible for certain emotions, and these emotions not only direct our behavior, but can be recognized reliably from facial expressions[42]. While this was the approach we took in this paper,[43] challenges the fundamental assumption of emotions having dedicated brain circuits and being recognizable from facial expressions in her popular theory of constructed emotion.[43] argues that emotions are constructed in the brain as the outcome of the complex interplay between different brain regions, and that part of emotions is learned, or the way emotions manifest can be impacted by learned information[43].

The theory of constructed emotion challenges the approach that we can reliably detect emotions simply from people's faces. What this means for our research is that future research could investigate alternative approaches for emotion detection (such as self-reports and bodily sensors) to boost the reliability of the findings. Furthermore, we encourage scholars to examine the fundamental assumptions they make in the study of emotions in the context of gamification and to see whether emotions are indeed ubiquitous or whether we see evidence of the manifestation of individual differences.

Finally, emotional responses should be further examined outside the controlled lab environment, such as classrooms or home-based e-learning contexts, as emotional engagement may differ in naturalistic educational settings, due to social interactions and external stressors[36,44]. While we acknowledge the challenges associated with this type of approach, such studies have the potential to yield unprecedented insights for the community by enabling an understanding of students' emotions in an ecological context.

Methods

Study Design

In this study, we investigated the effects of gamification on students' basic emotions. To achieve this, we conducted a between-subjects controlled experiment. By conducting this study, we aim to move beyond responses centered on students' subjective experiences and gain a clearer understanding of the effects of gamification on their emotions. Data collection was conducted in a usability laboratory between November and December 2023. The controlled lab environment enabled monitoring of physiological and emotional responses during system interaction and allowed for immediate post-task interviews. In this study, we did not aim to understand exactly which specific aspect of gamification causes a particular emotion.

Participants

Participants were recruited via the Tampere University DMLab pool, utilizing ORSEE3 software for coordination[45], an internal system at Tampere University used exclusively to invite students to participate in studies conducted by university researchers. Due to the interdisciplinary nature of the study, no prior exclusion criteria were used. After data collection, we filtered outliers with data imputation, applying the mean value for each row with a Z-score larger than 2.5. The final sample, excluding outliers, comprised 62 students from 19 countries; 33 identified themselves as female, 26 as male, two as non-binary, and one preferred not to respond. The average age was 25 years, with a standard deviation of 5.00 and a variance of 26.00. Participants had never used the platform before. Each participant received a movie ticket as compensation (i.e., each ticket cost 8 euros). None of the participants had prior contact with the gamified LMS used in the system or with the researchers involved in the study. The Institutional Review Board approved the study, and participants provided informed consent for participation and recording.

Materials

We employed the gamified educational system Eagle-edu (`https://eagle-edu.com.br/`) to conduct our study (i.e., present the educational tasks for the participants). We chose this system because of its adaptability, which empowers educators to create diverse educational activities. The system incorporates 21 distinct gamification elements, which align with the Taxonomy of Gamification Elements for Educational Environments (TGEEE) proposed by[46]. These elements can be selectively

enabled or disabled by teachers, allowing them to personalize the gamification designs. Thus, the system allowed us to assemble the appropriate gamification design for the study (i.e., for both experimental and control groups). At the same time, in addition to being used commercially, allowing it to imitate a real environment as closely as possible, the system has already been used in several scientific studies, *e.g.*,[24,47,48], thus avoiding biases related to the use of the system.

In our study, we utilized two different versions of the system: the version used with the participants in the experimental group presented a series of educational tasks incorporating a typical gamification design consisting of 10 gamification elements that created a meta-game interaction around the learning activity (distinct from, for instance, serious games or game-based learning). The 10 gamification elements were specifically selected because they enabled quick identification and interaction within the system during the controlled study period. Other elements available in the system (*e.g.*, narrative, storytelling, and ranking) require a longer usage period to be perceived and promote interaction. The version used with the participants in the control group presented the same educational tasks but without gamification.

For the experimental group, the gamification design comprised 10 gamification elements, aligned with the taxonomy proposed by Toda et al.[46]:

- **Acknowledgment**: This element provides feedback recognizing specific player actions. In our system, this was represented by student badges.
- **Chance**: Linked to randomness and probability, this element affects the likelihood of certain outcomes.
- **Competition**: Arising when multiple players compete for a shared goal.
- **Economy**: This element involves game transactions, including the monetization of values. Our system featured coins as currency for in-game purchases.
- **Imposed Choice**: Players had to make decisions to progress in the game. Our system offered random options to increase rewards.
- **Level**: Representing hierarchical progression, levels in our system were categorized as Bronze, Silver, Gold, Ruby, and Diamond.
- **Objective**: Goals, either quantifiable or spatial, guided players in the game. In our system, a quest tree structure represented the objectives.
- **Points**: Points measure user performance, with experience points (XP) serving as the unit of measurement.
- **Progression**: This element tracked player advancement through a progress bar integrated into the activity tree.
- **Stats**: Displaying a player's results and progress, this element provides visible information on all user progress data.

The objective of the present study was not to isolate the effects of individual gamification elements but rather to evaluate the emotional responses associated with a realistic gamification design composed of multiple interacting elements. Consequently, individual engagement with specific elements was not analyzed separately. Examples of the system's graphical user interface are provided in the supplementary materials.

Procedure

The study was conducted in three key steps: *i)* random assignment, *ii)* system interaction, and *iii)* emotions assessment. In the *first step*, participants were randomly assigned to either the experimental or control group. The randomization process was carried out with the participant (manually) immediately before the start of the experiment, after the student entered the laboratory, where participants were asked to turn a cup over. Inside that cup, there was a paper indicating whether the participant would be in the experimental or control group. The participant was not informed which group they would be in. In the *second step*, system interaction, participants engaged with the system for at least 30 minutes, completing educational tasks and exploring its features. In the *third step*, emotion assessment, data were collected during system usage of the software Noldus Observer XT (i.e., through equipment attached to the participants' bodies). Figure 3 summarize the study's procedure.

Measurement

In our study, we utilized Noldus Observer XT (`https://www.noldus.com/observer-xt-human`) and FaceReader (`https://www.noldus.com/facereader`) to collect and analyze participants' emotional responses. These tools were selected for their robust capabilities in behavioral coding and facial expression analysis, critical for accurately capturing and interpreting emotional data. Figure 4 exemplifies the tools' usage.

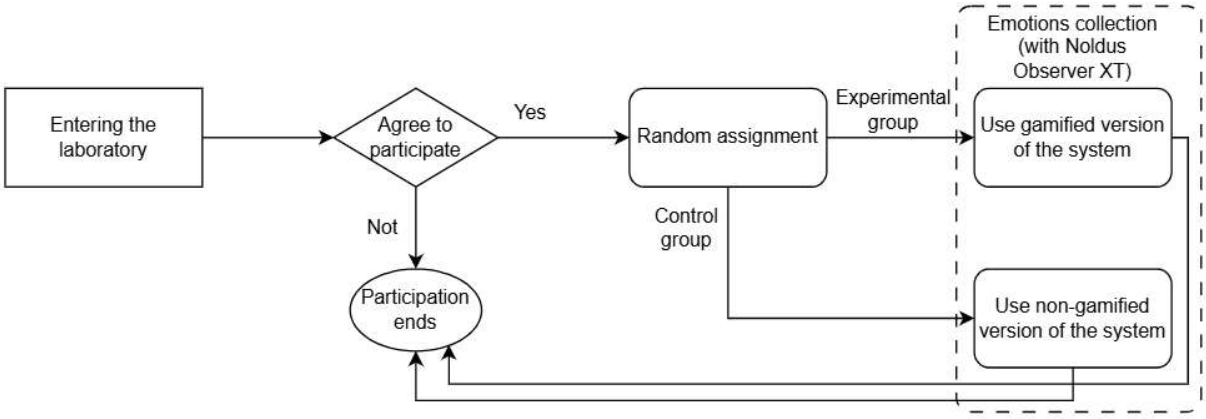


Figure 3. Study's procedure. The diagram illustrates the process of a study with two groups: an experimental group and a control group. Participants are randomly assigned to either group. Upon entering the laboratory, participants are asked if they agree to participate. If they do not agree, their participation ends. If they agree, those in the experimental group use a gamified version of the system, while those in the control group use a non-gamified version. Data collection for the students' emotions from both groups was conducted during the system usage.



Figure 4. View from a face captured by the camera inside a booth. The image was manipulated to preserve the participant's anonymity; however, it still showed how facial muscles were mapped.

Data analysis

To ensure the appropriate selection of statistical tests, we began by assessing the distribution of the collected data. Given the sample size of $N = 62$, we applied the Shapiro-Wilk test[49]. Developed by Shapiro and Wilk[49], this test is particularly suited for small to moderate sample sizes[19]. It computes a W statistic and *p-value* that compares the observed data to what would be expected under a normal distribution. A high *p-value*, i.e., $p > 0.05$, suggests that the data may be normally distributed[49].

Then, according to the Shapiro-Wilk test results (i.e., $p < .001$ for all emotions except neutral emotion p = .198), we conducted a Student's t-test[50] for normally distributed emotions' data (i.e., where data do not reject the null hypothesis of normality) and a non-parametric Mann-Whitney U test for the others (i.e., where data reject the null hypothesis of normality) to compare the means between the experimental and control groups, both using the two-sided approaches[51]. Because no statistically significant differences were observed across the tested emotions, correction for multiple comparisons would not affect the interpretation of the results.

Given the final sample size ($N = 62$), the study was primarily designed to identify relatively large differences between

conditions. As a result, the inferential analyses are more informative regarding the absence of large emotional effects than the absence of emotional effects altogether. Therefore, the findings should be interpreted with caution, particularly regarding potentially small but educationally meaningful effects that may require larger samples to detect reliably. The preregistration and study data are available in an OSF repository, which is provided as supplementary material for this article.

Ethical statements

The Institutional Review Board approved the study, and participants provided informed consent for participation and recording. All research procedures were conducted in strict adherence to the guidelines established by the Finnish National Board on Research Integrity (TENK).

Data availability

The authors declare that all data and materials are available as supplementary information in the article.

Supplementary material

The study dataset can be accessed from this link: https://doi.org/10.17605/OSF.IO/GTNHZ

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Author contributions statement

Wilk Oliveira: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources; Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision, and Project administration; Pasqueline Dantas Scaico: Conceptualization, Methodology, Investigation, Writing - Original Draft, and Writing - Review & Editing; Juho Hamari: Conceptualization, Methodology, Writing - Original Draft, Writing - Review & Editing, Supervision, and Funding acquisition; Samuli Laato: Writing - Original Draft; Alexandre Jun Hayasaka: Visualization, Formal analysis; Leonardo Tortoro Pereira: Visualization, Formal analysis, and Writing - Original Draft.

Notes

During the preparation of this work, the authors utilized generative artificial intelligence (i.e., Microsoft Copilot) to improve the grammatical quality of the text. After the utilization of this tool/service, the authors diligently reviewed and edited the content as necessary, assuming full responsibility for the publication's content. This article is an extension of the conference paper by[52]. This article is part of a broader research project examining the effects of gamification on multiple dimensions of the student experience. This project has already generated several publications, including[31,53].

Additional information

Competing interests: The first author is a partner at the company that granted the rights to use the system Eagle-edu free of charge for research purposes in this project.